

Foundation (Jalapeno)

Q1. What is the vertex of the graph of $y = |x - 2|$?

- (A) (0, 2)
 (B) (2, 0)
 (C) (2, 2)
 (D) (-2, 0)
 (E) (0, 0)

Q2. The graph of $y = |x + 1|$ has an axis of symmetry at:

- (A) $x = -1$
 (B) $x = 0$
 (C) $x = 1$
 (D) $y = -1$
 (E) $y = 0$

Q3. The absolute value function $y = |x|$ opens:

- (A) Up and down
 (B) Left and right
 (C) Only up
 (D) Only down
 (E) Not at all

Q4. A stock's price is $|x - 5|$ dollars, where x is the number of days since its initial public offering. On which day is the price 0?

- (A) Day 5
 (B) Day 10
 (C) Day 15
 (D) Day 0
 (E) Day 20

Q5. The tax on a stock trade is given by $y = |0.1x|$, where x is

the number of shares traded. What is the tax on 50 shares?

- (A) 5
 (B) 10
 (C) 0.5
 (D) 0.1
 (E) 0

Q6. Which of the following is the graph of $y = |x| + 2$?

- (A) A horizontal shift of the graph of $y = |x|$
 (B) A vertical shift of the graph of $y = |x|$
 (C) A reflection of the graph of $y = |x|$
 (D) A rotation of the graph of $y = |x|$
 (E) None of the above

Q7. A savings account earns interest given by $y = |0.05x|$, where x is the amount deposited. What is the interest earned on a

1000

deposit?

- (A) 50
 (B) 0.05
 (C) 0.5
 (D) 500
 (E) 0

Q8. The graph of $y = |x - 3| + 1$ has a minimum value of:

- (A) 0
 (B) 1
 (C) 2
 (D) 3
 (E) 4

Open Ended Questions (FRQ)

Q9. Graph the absolute value function $y = |x + 2| - 1$ and identify its vertex, axis of symmetry, and direction of opening. Required space: 5 lines

Q10. A company's profit is given by $y = |x - 10| - 5$, where x is the number of products sold. Find the number of products that must be sold for the company to break even. Show your work and explain your reasoning. Required space: 5 lines

Chef's Tips

- Emphasize the importance of understanding the properties of absolute value functions.
- Encourage students to use graphing calculators to visualize the graphs.
- Remind students to check their work and explain their reasoning.